

RESEARCH PROPOSAL

FAULT-TOLERANT CONTROL FOR SINGLE-PHASE DUAL ACTIVE BRIDGE USING ADJUSTED PHASE SHIFT MODE



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STATEMENT PAGE

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A handwritten signature in black ink, appearing to be 'Krishna Laksheta', written in a cursive style.

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TABLE OF CONTENTS

TITLE PAGE	1
TABLE OF CONTENTS	1
LIST OF TABLES	3
LIST OF FIGURES	4
LIST OF SYMBOLS AND ABBREVIATIONS.....	5
ABSTRACT.....	6
CHAPTER I INTRODUCTION	1
1.1 Background	1
1.2 Problem Formulation	2
1.3 Research Objectives	3
1.4 Research Limitations	3
1.5 Research Benefit.....	3
1.6 Research Authenticity	3
CHAPTER II LITERATURE REVIEW AND THEORETICAL FOUNDATION..	6
2.1 Literature Review.....	6
2.1.1 Redudancy Based	6
2.1.2 Reconfiguration.....	7
2.1.3 Advanced Control.....	8
2.1.4 Proposed Approach.....	9
2.2 Theoretical Foundation	10
2.2.1 Dual Active Bridge	10
2.2.1.1 Single Phase Shift	11
2.2.2 Fault Types in Power Converters	11
2.2.2.1 Open-Circuit Faults	11
2.2.2.2 Short-Circuit Faults	12
2.3 Hypotheses	13
CHAPTER III RESEARCH METHODS	14
3.1 Research Tools	14
3.1.1 Laptop Specifications	14
3.1.2 Software.....	14
3.1.3 DAB Converter	14
3.1.4 LAUNCHXL-F28379D Microcontroller.....	14
3.2 Materials.....	15
3.2.1 Simulation Model Data	15
3.2.2 Hardware Verification Data.....	16
3.3 Research Method	16
3.3.1 Fault-Tolerant Control Algorithm Development	16
3.3.2 Microcontroller Implementation	18
3.3.3 Hardware Testing and Verification.....	19

3.3.4 Result Analysis	20
CHAPTER IV RESEARCH SCHEDULE	21
REFERENCES	22
APPENDIX	L-1
Proof of Submission.....	L-1
Supervisor's Approval	L-2

LIST OF TABLES

Table 1.1	Research in Fault-Tolerant Control Methods for OCF Detection	4
Table 3.1	Specifications of Dual Active Bridge Converter	15
Table 3.2	Specifications of LAUNCHXL-F28379D Microcontroller [40].....	15
Table 4.1	Research Activity Plan Schedule	21

LIST OF FIGURES

Figure 2.1	Dual Active Bridge Topology	10
Figure 3.1	LAUNCHXL-F28379D Microcontroller	15
Figure 3.2	Flowchart of the Research	17
Figure 3.3	Flowchart of the Fault-Tolerant Control Algorithm.....	18
Figure 3.4	Flowchart of the Microcontroller Implementation	19
Figure 3.5	Flowchart of the Hardware Implementation	20

LIST OF SYMBOLS AND ABBREVIATIONS

DAB	Dual active bridge
OCF	Open-circuit fault
SCF	Short-circuit faults
FTC	Fault-tolerant control
SPS	Single-phase shift
APS	Adjusted phase shift
dc	Direct current
ac	Alternating current
ZVS	Zero-voltage switching
ANN	Artificial neural network
IGBT	Insulated Gate Bipolar Transistors
IDE	Integrated development environment
MOSFET	Metal-oxide-semiconductor field-effect transistor
PWM	Pulse width modulation

ABSTRACT

The dual active bridge (DAB) converter is a widely used power conversion topology known for its bidirectional power flow and galvanic isolation. However, its reliability is compromised when faults occur, especially open-circuit faults (OCFs) that can disrupt power transfer and damage components. While many fault-tolerant control (FTC) strategies exist, most require additional hardware, increasing complexity and cost, or involve computationally expensive techniques. This research proposes a fault-tolerant control system for a single-phase DAB converter that addresses open-circuit faults without requiring significant hardware modifications or reconfiguration. The system utilizes a single-phase shift (SPS) in the pre-fault mode, transitioning to an adjusted phase shift (APS) mode during fault conditions. It handles multiple faulty switches using only one current sensor on the primary side, reducing sensor requirements and simplifying the system design. The proposed approach minimizes computational power, offers cost-effectiveness, and allows easy integration with microcontrollers for hardware verification. This research contributes to simplifying fault-tolerant control in DAB converters while improving reliability and efficiency, making it a valuable solution for practical applications in electric vehicle charging, energy storage systems, and renewable energy integration.

Keywords—Fault-tolerant control, Dual active bridge converter, open-circuit faults, phase shift, microcontroller integration.

CHAPTER I

INTRODUCTION

1.1 Background

The dual active bridge (DAB) converter is an advanced dc-dc converter recognized for its ability to facilitate bidirectional power flow while also providing galvanic isolation [1], [2]. This topology incorporates two active H-bridges connected through a high-frequency transformer, making it capable of efficiently converting power across a broad voltage range [3]. A key feature of the DAB is its ability to achieve zero-voltage switching (ZVS), significantly reducing switching losses during operation, thus improving overall efficiency [4], [5]. The DAB is gaining increasing significance in practical applications, especially in contexts where rapid energy transfer is essential. It is widely used in electric vehicle charging systems [6], energy storage interfaces [7], and renewable energy integration within microgrid frameworks [8]. These attributes make the DAB a preferred choice for high-power applications, enabling higher power density and operational flexibility due to its ability to operate efficiently over a broad power range [1].

Researchers in [9] conducted a survey revealing that 76% of converter failures are attributed to semiconductor devices (57%) and gate drivers (19%). These failures are characterized by open-circuit faults (OCFs) and short-circuit faults (SCFs) [10], [11]. OCFs occur when one or more switches fail to conduct, which results in a loss of power transfer capability, dc bias in transformer currents, and potential component damage [10], [11]. In contrast, SCFs arise when a switch or load becomes shorted, leading to excessive current flow, which can cause overheating and component failure [12].

Meanwhile, the DAB converter consists of multiple switches that are critical to its operation. Each switch within the DAB system is a potential point of failure. Given that the failure of any switch can lead to significant disruptions in power transfer, it can also cause potential damage to other components. The reliability of the switch system in a DAB converter becomes a crucial consideration. While DAB converters offer significant advantages, the presence of multiple switches introduces a higher risk of faults. Therefore, it is crucial to develop fault-tolerant control (FTC) mechanisms to ensure the DAB converter can operate safely and efficiently, even in the presence of faults.

Several studies have proposed various approaches to mitigate SCFs in DAB converters. Research in [13], [14], [15], [16] has developed FTC methods, such as adding series capacitors and a fast-acting fuse [13], incorporating a voltage-doubler rectifier [15], and employing sensorless fault detection [16]. Despite the destructive nature of SCFs, protection schemes have become standardized and are now integrated into gate driving circuits for industrial applications [17], [18], [19], [20], [21]. However, given the extensive research and established protection mechanisms already in place for SCFs, further development in this area is less critical.

With significant advancements in SCF protection, current research increasingly

focuses on open-circuit FTC in DAB. OCFs in switches are particularly challenging to manage due to the lack of standardized design guidelines and their higher occurrence rate compared to SCFs [22]. Much research has been dedicated to addressing OCFs as a primary objective. For instance, [13] and [23] employ redundancy components such as capacitors, fuses, and inductors. In contrast, [24] propose reconfiguration techniques, switching the DAB converter topology from full-bridge to half-bridge to mitigate faults. Meanwhile, [25] suggests altering the transformer topology to enhance fault tolerance. Approaches that rely on sensors, such as voltage and current monitoring, have been explored in works by [24], [26], [27], and [28]. Moreover, advanced control strategies, including vector-based analysis and disturbance monitoring, are utilized in [9] and [29]. Additionally, artificial neural network (ANN) has been employed by [30] for OCF detection, offering high accuracy.

Despite significant advancements in fault-tolerant strategies for DAB converters, several limitations remain. Many existing approaches rely on additional hardware components, such as redundancy elements (capacitors, fuses, inductors), which increase system complexity and cost [13], [23]. Furthermore, methods that employ reconfiguration or advanced control techniques often demand significant computational power, which can result in slower response times and higher energy consumption [9], [24], [29], [30]. Additionally, while sensor-based fault detection strategies are widely used, they are not always scalable, and the dependency on multiple sensors can increase system cost and complexity [26], [27], [28]. Some techniques also face limitations in adapting to multiple fault conditions without requiring extensive rewiring or system adjustments [25]. Therefore, a solution that addresses these challenges is of critical importance.

This research proposes a fault-tolerant control system for a single-phase DAB converter that addresses open-circuit faults without requiring significant hardware modifications or reconfiguration. The system operates with an single-phase shift (SPS) in the pre-fault mode, transitioning to an adjusted phase shift (APS) mode under fault conditions. It can handle multiple faulty switches with mitigation strategies, using just one current sensor on the primary side, which reduces sensor requirements and simplifies the design. The method minimizes computational power, is cost-effective, and integrates easily with microcontrollers for hardware verification.

1.2 Problem Formulation

Despite advancements in fault-tolerant strategies for DAB converters, significant challenges persist. Many existing methods require additional hardware components, such as redundancy elements, which increase system complexity and cost. Furthermore, approaches that involve reconfiguration or advanced control techniques often demand substantial computational power, resulting in slower response times and higher energy consumption. Sensor-based fault detection, while commonly used, adds to the system's complexity and cost due to the dependency on multiple sensors. Additionally, many techniques struggle to adapt to multiple fault conditions without requiring extensive rewiring or adjustments to the system.

1.3 Research Objectives

1. Develop a FTC system for a single-phase DAB converter to address OCFs.
2. Integrate the FTC system with a microcontroller for real-time hardware verification.
3. Minimize additional hardware requirements and overall system complexity.
4. Reduce computational load to enable efficient embedded implementation.
5. Achieve cost-effective and efficient system operation under fault scenarios.

1.4 Research Limitations

The research will be conducted through software simulation, followed by hardware verification at the laboratory level. The hardware verification will be limited to a controlled laboratory setup. The FTC system will also be explicitly designed for a single-phase DAB converter. The study will also be limited to the detection and mitigation of OCFs.

1.5 Research Benefit

This research aims to advance FTC strategies in power electronic devices with a focus on DAB converters. The objective is to develop a more efficient and cost-effective fault-tolerant system that enhances the reliability and performance of DAB converters. This advancement is expected to contribute significantly to both academic research and industrial applications.

From a societal standpoint, implementing this research can lead to more reliable power conversion systems, minimizing downtime and increasing energy system efficiency. These improvements can result in significant cost savings and enhanced energy security for various sectors. Furthermore, the insights gained may align with and support national efforts to strengthen the sustainability and resilience of energy infrastructure.

1.6 Research Authenticity

Extensive research has been conducted in the field of FTC to address OCFs. These studies aim to develop solutions or refine existing methods to overcome challenges identified in previous works. Despite these ongoing efforts, many current approaches still face significant limitations, including high complexity, increased costs, challenges with real-time applicability, and reduced robustness under varying operational conditions.

Table 1.1 summarizes the methods proposed in recent literature, with a focus on their advantages and limitations. The table provides a foundation for understanding the necessity of the current study, which aims to address the identified gaps by introducing a novel approach designed to improve performance and feasibility in real-time applications.

In [13], a fault-tolerant scheme is proposed for detecting both OCFs and SCFs using desaturation detection. This method ensures fast fault detection but introduces increased hardware complexity due to the need for multiple logic circuits. Similarly, [23] presents an approach using a blocking capacitor and modified switching scheme, allowing the converter to operate at reduced power following fault detection. While avoiding addi-

Table 1.1. Research in Fault-Tolerant Control Methods for OCF Detection

References	Year	Methods	Limitation
[13]	2022	Desaturation detection for OCFs and SCFs	Requires eight logic circuits, increasing hardware complexity and costs
[23]	2023	Blocking capacitor with switching scheme modification	Dependent on load conditions and software complexity; not suitable for real-time systems
[24]	2022	Voltage sensor per bridge leg	Noise sensitivity reduces robustness in noisy environments
[25]	2023	Transformer reconfiguration with center-tap current monitoring	Requires structural modifications to the converter hardware, which may not be viable for all applications
[27]	2021	Current sensors on each phase, logic-based fault detection	Susceptible to false detections due to noise sensitivity
[26]	2023	Improved logic-based fault detection	Noise sensitivity persists
[28]	2023	Asymmetric resonant half-bridge converter with auxiliary switches	Significant software complexity, challenging for real-time applications
[9]	2024	Vector-based detection approach using dc offset relationships	Requires additional low-bandwidth current sensors, increasing sensing overhead and system complexity
[29]	2022	Gate voltage pulse disturbance analysis	Requires high-resolution measurement tools, increasing complexity
[30]	2022	ANN for fault diagnosis	Requires multiple sensors and significant computational resources, increasing costs and complexity
This research	2025	APS with one current sensor	

tional sensors, this method depends on load conditions and introduces software complexity, making it less suitable for real-time applications.

Meanwhile, [24] offers a more straightforward detection process by using just one voltage sensor per bridge leg to monitor switches for OCFs. This method reduces the number of sensors but compromises robustness in noisy environments, limiting its effec-

tiveness in applications where noise is a concern.

A different approach by [25] tackles the challenge of hardware additions by proposing an FTC method that reconfigures the Transformer with center-tap current monitoring. This solution detects OCFs in DAB converters without requiring extra sensors. Although it enables fast, load-independent detection, the method demands structural modifications to the converter hardware, which may not be viable for all applications.

Research in [27] and [26] utilizes current sensors on each phase for OCF detection, which provides improved fault coverage. However, both methods are noise-sensitive, with [27] being particularly prone to false detections. Likewise, [28] proposes an OCF detection method for asymmetric resonant half-bridge converters using minimal voltage sensors. However, it introduces significant software complexity, posing challenges for real-time applications.

A more advanced control technique is introduced in [9], which utilizes a vector-based detection approach based on dc offset relationships in three-phase current signals to improve noise immunity. This method enhances fault detection performance, yet it requires additional low-bandwidth current sensors on the transformer side, increasing sensing overhead and complicating the system.

Similarly, the detection process is enhanced by analyzing disturbances in gate voltage pulses, as demonstrated in [29]. This method identifies OCFs within a single switching cycle and enables faster fault diagnosis. However, it relies on precise gate signal analysis, necessitating high-resolution measurement tools, further increasing the system's complexity.

An unexpected approach is presented in [30], where an ANN is used for fault diagnosis and achieves an impressive accuracy of up to 99.82% in detecting OCFs. This approach is truly remarkable, as it demonstrates the power of machine learning in a domain where traditional methods have been more common. However, despite its outstanding performance, this method requires multiple sensors and significant computational resources, resulting in higher implementation costs and increased system complexity.

This research contributes a novel FTC system for a single-phase DAB converter that addresses OCFs more efficiently and practically. Unlike previous studies, this method does not require adding complex components, using multiple sensors, or modifying transformer topologies. Instead, it operates with a single current sensor on the primary side, reducing hardware complexity and costs. The system transitions from a pre-fault SPS mode to an APS mode under fault conditions and can handle up to four faulty switches (two on each side of the converter). By minimizing sensor requirements, computational power, and the need for hardware modifications, the proposed method offers a more straightforward and cost-effective solution. It can easily integrate with microcontrollers for real-time applications and serves as an essential addition to the field.

CHAPTER II

LITERATURE REVIEW AND THEORETICAL FOUNDATION

2.1 Literature Review

Significant research has focused on addressing open-circuit faults in dual active bridge (DAB) converters in recent years. Various methods have been proposed to effectively detect and mitigate the impact of these faults, including redundancy-based, reconfiguration, and advanced control. These solutions enhance the converter's reliability and robustness and ensure optimal performance under fault conditions.

2.1.1 Redudancy Based

In power electronics, redundancy-based fault tolerance involves the use of backup components or systems that take over the functionality of a failed component. This approach enhances the reliability and robustness of systems. By incorporating additional components such as backup switches or fuses, redundancy helps maintain system performance and minimize the impact of faults.

One approach, presented in [23], employs a blocking capacitor in combination with a modified switching scheme. This adjustment allows the converter to continue operating at half of its rated power after an open-circuit fault (OCF) is detected. A notable advantage of this method is that it avoids additional sensors and simplifies the design. However, the method is highly dependent on load conditions, and its complexity increases due to the software adjustments required. As a result, it becomes less suitable for real-time systems. Additionally, the inclusion of a blocking capacitor raises the system's cost, which could be a limiting factor for some applications.

The method in [31] uses a redundant leg with multiple fuses to enhance the reliability of DAB converters. This method isolates a faulty leg using redundancy and ensures that the system continues operating even when one component fails. However, additional components are required to disconnect the faulty device and integrate the redundant component into the circuit to implement this fault-tolerant operation. Furthermore, a separate fault isolation unit is needed to isolate the damaged switch. While this approach improves the system's fault tolerance, it introduces additional complexity and increases the overall cost of the system, making it a less attractive solution for cost-sensitive applications.

In [32], the concept of using a redundant winding for fault-tolerant operation is explored. This method involves connecting fuses in series to isolate the fault, which helps to ensure continued operation in case of a failure. While this method allows for fault correction on the primary side of the converter, it comes with a trade-off in terms of efficiency. Moreover, the use of multiple fuses, windings, and relays increases the cost of the overall system. While effective for limited fault scenarios, this method does not provide comprehensive protection against all types of faults, which could limit its application in more critical systems.

The research in [33] proposed a solution that employs pre-charged backup switches activated immediately during OCFs in the IGBTs for DAB converters. This method ensures that power flow remains uninterrupted, with minimal disruption to the system. Simulation results demonstrate that the approach offers a rapid fault recovery time of just $6\mu\text{s}$ and maintains system efficiency above 95% during fault-tolerant operation, significantly reducing power losses compared to traditional methods. However, the study acknowledges that implementing this strategy could be costly and complex. It presents challenges in terms of design and cost-effectiveness in specific applications.

2.1.2 Reconfiguration

Reconfiguration-based fault tolerance is a strategy that allows systems to maintain functionality by dynamically adjusting their structure in response to faults. In contrast to redundancy, which uses backup components, reconfiguration involves modifying the system's configuration or topology to bypass the faulty components or adapt the system's operation. This method aims to minimize the impact of failures and maintain system performance, often by re-routing power or reassigning tasks to healthy components.

In [13], a fault-tolerant scheme using desaturation detection is proposed to identify both open-circuit faults (OCFs) and short-circuit faults (SCFs) in DAB converters. The scheme compares the collector voltage with gate drive signals via logic gates, achieving fault detection within $5\mu\text{s}$. This quick detection enhances the converter's ability to respond to faults in real time. This method reconfigures the converter to adjust its operation mode after detecting a fault. However, implementing this method requires eight logic circuits, which increases both hardware complexity and costs. This makes the method less ideal for low-cost applications.

The work in [24] introduces a fault-tolerant DAB converter designed for dc distribution systems, which aims to maintain uninterrupted power transfer even after multiple OCFs in semiconductor switches. The proposed solution uses a reconfigurable topology with a center-tapped transformer, two auxiliary inductors, and only four additional voltage sensors. This enables the converter to seamlessly switch between normal and post-fault operation modes. This approach balances performance and cost by minimizing the need for additional components and complexity. They also test the proposed method with experimental results using a 250 W prototype, showing that the converter can continue power transmission under single and dual OCF conditions while minimizing stress during transitions between operational modes.

The method presented in [25] introduces an OCF detection technique for DAB converters based on center-tap current monitoring of the transformer. This approach enables fast and load-independent detection of OCFs. It significantly improves over traditional methods that typically rely on the output load. However, implementing this method requires modifications to the converter's hardware, which may complicate the design and increase costs.

The method proposed in [27] detects and isolates OCFs in switches by plac-

ing current sensors on each phase. When a fault occurs, the method reconfigures the gate-driving circuit of the DAB converter to establish a conducting path. This path enables the gate activation current to pass through the gate resistors and generates a short-duration pulse voltage. This logic-based fault detection method only requires current sensing on one side of the transformer. While effective, the approach is highly sensitive to noise, making it susceptible to false fault detections due to noise interference.

The method in [27] is further improved in [26], where enhancements are made to the logic-based fault detection technique. While the updated approach refines the detection process, it still relies on current sensing on one side of the transformer. Despite these improvements, the method remains susceptible to noise sensitivity, which can lead to false fault detections, limiting its effectiveness in noisy environments.

Research in [28] presents a detection strategy for OCFs in an asymmetric resonant half-bridge converter. The method identifies faults by monitoring the voltage between the two converter legs and compensates for the fault using two auxiliary switches. The system reconfigures itself upon fault detection by activating these auxiliary switches to maintain operation. While the strategy operates independently of the output load and minimizes sensor usage, it introduces significant complexity in software implementation. This complexity could hinder its real-time application in fault detection systems.

2.1.3 Advanced Control

Advanced control strategies rely on algorithms and control techniques rather than redundancy components or alterations to the system topology. These strategies focus on enhancing the fault tolerance of systems by dynamically adjusting operational parameters in real time. Advanced control methods use sophisticated control algorithms to ensure continued operation and optimal performance under fault conditions without additional hardware. They provide a more flexible and efficient solution to fault management.

One advanced control technique is dual-loop control, which combines voltage and current envelope control to suppress oscillations and ensure stable operation during faults [34]. While effective in maintaining system stability, this control system may lead to higher costs and more complicated system integration. These factors could limit its applicability in specific contexts. Nonetheless, dual-loop control is crucial for maintaining uninterrupted operation, even when faults occur in one or more components.

Research in [29] refines the detection method by analyzing disturbances in gate voltage pulses to identify OCFs in switches. This technique enables fault diagnosis within a single switching cycle, significantly improving response time. However, its reliance on precise gate voltage measurements may introduce challenges in certain operating conditions, particularly in noisy environments.

In [9], a vector-based method is proposed to enhance noise immunity in fault diagnosis. This method identifies faults by analyzing three-phase currents' dc offset vector relationships. Despite its effectiveness, the approach requires additional low-bandwidth current sensors on the transformer side, which adds to the system's complexity and cost.

A recent approach by [30] employs artificial neural network (ANN) for OCF detection, offering high speed and accuracy, with diagnostic performance reaching up to 99.82%. Despite its promising results, this method is costly and complex to implement due to the need for multiple sensors and significant hardware and software requirements. This makes it less feasible for low-cost applications.

Furthermore, specific modulation schemes have been designed to sustain power delivery and prevent voltage spikes. These schemes remain effective even in cases where multiple switches or capacitors fail [35]. Despite these advantages, these techniques still face challenges in real-time fault detection, particularly in high-power applications, where faster response times are critical.

2.1.4 Proposed Approach

The proposed fault-tolerant control (FTC) system for the single-phase DAB converter provides a more efficient and practical solution than previous methods. Unlike the approach in [23], which requires a blocking capacitor and a modified switching scheme, the proposed system does not rely on complex components. As a result, it reduces both cost and hardware complexity. Additionally, while methods like those in [32], [31], and [33] rely on redundant components, the proposed method eliminates the need for such components.

The proposed method reconfigures the system from a pre-fault single-phase shift (SPS) mode to an adjusted phase shift (APS) mode when a fault occurs. This approach reduces both complexity and cost compared to [13], [27], and [26], which rely on additional components and logic circuits for fault detection and isolation. Additionally, this method does not require a redesign of the transformer, unlike the approach in [24] and [25], which necessitate modifications to the transformer. It also does not rely on the output load, as seen in [28]. This characteristic allows the system to offer greater flexibility and efficiency.

Compared to methods like the dual-loop control in [34], which increases cost and complexity due to additional hardware, the proposed approach offers a more straightforward, more cost-effective solution by relying on a single current sensor and micro-controller integration. While techniques such as the ANN-based method in [30] and the vector-based method in [9] provide accurate fault detection, they require multiple sensors, complex hardware, and significant software resources. In contrast, the proposed method minimizes the need for additional hardware and sensors, reducing cost and complexity. Similarly, approaches like those in [29] and [35] enhance fault detection but still face challenges in real-time operation, particularly in high-power applications. The proposed method simplifies fault detection and system operation and ensures efficient performance with minimal susceptibility to noise. This makes real-time implementation with micro-controllers more feasible.

In conclusion, the proposed FTC system significantly improves redundancy-based, reconfiguration, and advanced control methods. It reduces hardware complexity, sensor

requirements, and costs while improving fault tolerance. Its implementation with a micro-controller for real-time fault detection makes it a flexible, scalable, and cost-effective solution for fault-tolerant DAB converters, ensuring reliable operation even in complex and fault-prone environments.

2.2 Theoretical Foundation

2.2.1 Dual Active Bridge

The DAB converter is an advanced type of dc-dc converter known for its versatility and efficiency in facilitating bidirectional power flow while providing galvanic isolation [1], [2]. It operates with a high-frequency transformer, central to its operation, enabling efficient power conversion across a wide voltage range. As shown in Figure 2.1, the DAB converter consists of two active H-bridges connected through a transformer, with each bridge capable of controlling the voltage and current flow in both directions. This topology offers significant benefits in terms of power density, reliability, and efficiency, making it a popular choice in many high-power applications [3].

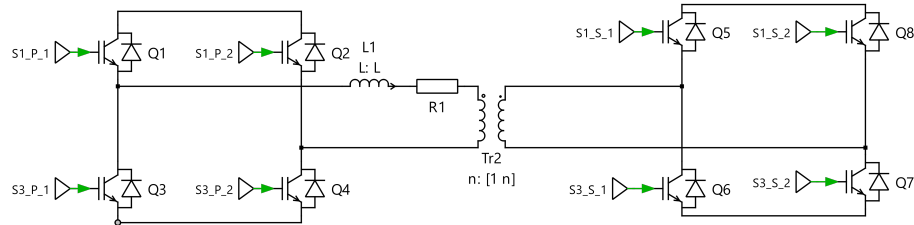


Figure 2.1. Dual Active Bridge Topology

This configuration allows bidirectional power flow, enabling the converter to transfer energy efficiently while maintaining zero-voltage switching (ZVS). ZVS significantly reduces switching losses by ensuring the voltage across the switches is zero when turning on or off. As a result, the reduction in switching losses improves the overall efficiency of the converter [4], [5]. The power transfer, P , between the two bridges is regulated by controlling the phase shift ϕ between V_{primary} and $V_{\text{secondary}}$ and is given by the equation

$$P = \frac{nV_iV_o\phi(\pi - \phi)}{2\pi^2 f_s L_r}, \quad (1)$$

where f_s is the switching frequency, L_r is the external series inductor of the DAB, and n is the turns ratio of the transformer [13]. In Equation 1, the power transfer P is regulated by the phase shift ϕ , which is the difference in phase between the primary and secondary voltages. V_i represents the input voltage on the primary side, while V_o is the output voltage on the secondary side. The term n refers to the transformer's turn ratio, which affects the voltage scaling between the primary and secondary sides. The switching frequency, denoted as f_s , determines the timing of the switching events. Meanwhile, L_r represents the external series inductor that controls the current flow and energy storage during operation.

In practical applications, the DAB converter has gained significant importance, especially in systems where high efficiency and fast energy transfer are critical. Its ability

to manage bidirectional power flow, coupled with its ZVS capability, makes it particularly well-suited for applications such as electric vehicle charging systems, where efficient power transfer between the grid and the vehicle is essential [6]. Moreover, DAB converters are increasingly utilized in energy storage interfaces [7] and in the integration of renewable energy sources within microgrid frameworks [8]. These converters provide reliable and efficient solutions for energy conversion in dynamic and distributed power systems.

2.2.1.1 Single Phase Shift

The SPS control strategy is one of the most commonly used methods for regulating the power transfer in a DAB converter [36]. This control approach involves adjusting the phase shift between the voltage waveforms of the primary and secondary bridges. By modulating the phase shift, the power transfer between the primary and secondary sides of the converter can be controlled, which in turn regulates the output power delivered to the load.

To express this relationship mathematically, the power transfer, P , is determined by the phase shift between the primary and secondary voltages. The Equation 2 is derived from the general power transfer Equation 1 by applying the approximation $\sin(\phi) \approx \phi$ for small phase shifts [37]. By applying the aforementioned simplifications, the power transferred by the single-phase DAB converter is given by

$$P = \frac{V_i V_o \cdot n}{2f_s L_r} \cdot \sin(\Delta\phi), \quad (2)$$

where V_i is the input voltage, V_o is the output voltage, n is the turns ratio of the transformer, f_s is the switching frequency, L_r is the external series inductor, and $\Delta\phi$ is the phase shift between the primary and secondary voltages.

The highest power transmission occurs when the phase shift reaches its maximum value,

$$\phi = \pm \frac{\pi}{2}, \quad (3)$$

as noted in [38].

2.2.2 Fault Types in Power Converters

2.2.2.1 Open-Circuit Faults

The occurrence of an OCF disrupts the normal conduction process, leading to a loss in power transfer capability. This fault type can lead to the development of dc bias in transformer currents, reduced efficiency, and the potential for damage to components over time [10], [11]. While OCFs are generally not catastrophic, they can still cause significant issues, as the converter may continue to operate at a reduced performance level. However, the prolonged current or voltage stress resulting from an OCF can gradually degrade the switches and connected components. This degradation could cause a system failure in the long term.

The causes of OCFs are varied and can include issues such as bond wire lift-off or rupture, driver unit malfunctioning, cooling system failure, or control unit faults. Additionally, failures in auxiliary circuits can lead to overcurrent, overvoltage, and temperature-related problems, further exacerbating the fault condition and increasing the risk of damage [13]. While the converter may continue to operate despite the fault, the increased stress on the system can reduce overall reliability and efficiency.

One of the primary challenges associated with OCFs is the absence of a standardized detection or mitigation design. The impact of an OCF can vary significantly depending on the converter topology, making it essential to develop topology-specific fault detection and management strategies. Understanding the behavior and impact of OCFs in different converter configurations is critical to prevent further damage. The failure of a switch to conduct leads to an increase in current stress and the introduction of dc bias current, both of which contribute to higher power losses and a greater likelihood of damaging other components within the system [24].

In conclusion, while OCFs allow the converter to continue operating under steady-state conditions, they introduce voltage imbalances and increase stress on the system, which can affect long-term performance and reliability [25]. Therefore, it is essential to develop effective fault detection and mitigation techniques specific to each converter topology to ensure optimal performance and prevent catastrophic failure [9].

2.2.2.2 Short-Circuit Faults

SCFs occur when two or more terminals are unintentionally connected, creating a path of low resistance that results in an excessive current flow. This phenomenon can lead to catastrophic damage to the power converter, often resulting in the overheating of components, the destruction of semiconductor devices, and potential system failure [13]. Due to the severity of SCFs, they are considered highly critical and must be isolated immediately upon detection to prevent further damage to other components within the system.

SCFs typically arise from causes such as bond wire failures, driver unit malfunctions, cooling system issues, or auxiliary circuit failures [13]. Once an SCF occurs, the resulting high overcurrents can create catastrophic conditions within the converter, particularly when these faults involve the semiconductor switches. The excessive current can cause shoot-through, where both the high and low-side switches conduct simultaneously, creating a direct short across the power supply and causing significant damage [24]. Therefore, it is essential to quickly detect SCFs to ensure timely isolation and prevent damage propagation throughout the converter.

Regarding the DAB converter, SCFs inevitably lead to catastrophic shoot-through and large short-circuit currents, which can seriously damage the semiconductor switches and other components. These faults must be immediately addressed to prevent further degradation of the converter's performance and to ensure the protection of other connected components [24]. For this reason, high-speed detection systems integrated into the gate

driving circuit are a standard design feature in modern converters to handle SCFs efficiently.

2.3 Hypotheses

This study aims to investigate the effectiveness of the proposed FTC system for handling OCFs in a single-phase DAB converter. The following hypotheses are put forward based on the research objectives and proposed method:

1. The APS mode can effectively manage up to four open-circuit faults (two on the primary side and two on the secondary side) in a single-phase DAB converter without additional components or complex hardware modifications. This FTC system will maintain the converter's stable operation while minimizing the faults' impact on its performance.
2. The proposed FTC system can be successfully implemented using a microcontroller platform and tested on actual DAB converter hardware. The system's real-time capability, operational efficiency, and reduced hardware complexity will demonstrate the feasibility of integrating the control mechanism into a practical solution. This solution will also be cost-effective for real-time applications.

These hypotheses will be tested by evaluating the performance of the proposed system under fault conditions and comparing the results with traditional methods in terms of system stability, efficiency, and hardware complexity.

CHAPTER III

RESEARCH METHODS

3.1 Research Tools

The research will utilize the following tools and equipment to design, simulate, and verify the fault-tolerant control (FTC) system for the single-phase dual active bridge (DAB) converter:

3.1.1 Laptop Specifications

The primary computing tool for this research will be a laptop with the following specifications:

- Processor: Intel(R) Core(TM) i5-7200U CPU @ 2.50GHz (4 CPUs)
- Memory: 8 GB RAM

These specifications will support running simulation software, environment programming, and algorithm development resource-efficiently.

3.1.2 Software

The research will make extensive use of the following software tools for simulation, programming, and control system development:

- **PLECS:** This simulation software will model the single-phase DAB converter, including its FTC system. PLECS will validate the FTC algorithm before its implementation in hardware. This validation ensures that the system behaves correctly under both normal and fault conditions during the simulation phase.
- **Visual Studio Code:** This integrated development environment (IDE) will be employed for programming in both C and Python. C will be used for hardware programming and simulation scripts related to control system development, while Python will be utilized for scripting simulations and processing results.

3.1.3 DAB Converter

The research will involve hardware verification using a DAB converter. This converter will be used to validate the performance of the proposed FTC system in real-world conditions. The detailed specifications of the DAB converter are summarized in Table 3.1.

3.1.4 LAUNCHXL-F28379D Microcontroller

The LAUNCHXL-F28379D microcontroller will program and implement the proposed FTC method for the DAB converter. It will handle the control algorithms and fault detection, which will be tested on the hardware setup. The detailed specifications of the LAUNCHXL-F28379D are summarized in Table 3.2. Additionally, Figure 3.1 shows the LAUNCHXL-F28379D microcontroller.

Table 3.1. Specifications of Dual Active Bridge Converter

Specification	Value
V_i, V_o	780 V
Power	6 kW
Transformer Ratio	1:1
Inductance	97.34 mF
MOSFET (metal-oxide-semiconductor field-effect transistor)	IMZ120R030M1HXKSA

Table 3.2. Specifications of LAUNCHXL-F28379D Microcontroller [40]

Specifications
Dual 200MHz C28x cores
1MB Flash and 204KB RAM
Trigonometric Math Unit (TMU)
4 ADCS with selectable 16-bit/12-bit operation
Serial Communications Peripherals: SCI, SPI, I2C, CAN, USB
2 User LEDs
40 GPIO

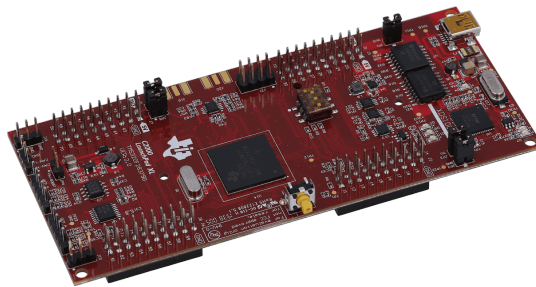


Figure 3.1. LAUNCHXL-F28379D Microcontroller

3.2 Materials

The data used in this research is primarily obtained from two sources: simulation models and hardware verification.

3.2.1 Simulation Model Data

The simulation data is collected from performance evaluations of the FTC algorithm implemented in the simulation model of the DAB converter. The key performance metrics include:

- **Execution Time:** The algorithm's time to perform fault detection and mitigation tasks.
- **Resilience:** The ability of the algorithm to maintain stable performance during fault conditions.
- **Fault Detection Accuracy:** The accuracy of detecting open-circuit faults (OCFs).
- **Control Performance:** Metrics such as the stability and regulation of voltage and current under fault conditions.
- **Computational Load:** The computational effort required to run the algorithm in real time.

These simulation results are obtained using the PLECS software, and the data is processed and analyzed using Python for further performance evaluation and comparison.

3.2.2 Hardware Verification Data

Data is collected from the DAB converter during testing for hardware verification. The key data points include:

- **Voltage Output:** The converter's output voltage is monitored to ensure correct operation.
- **Primary Side Current:** The current flows through the DAB converter's primary side.
- **Primary AC Voltage:** The ac voltage on the primary side.
- **Secondary AC Voltage:** The ac voltage on the secondary side.
- **Load Current:** The current consumed by the load connected to the DAB converter.
- **PWM (pulse width modulation) Signals:** PWM signals for both the primary and secondary sides.

These data are collected using an oscilloscope with voltage probes for voltage measurements and current probes for current measurements. The data is then processed to assess the converter's performance and verify the accuracy and robustness of the control system.

3.3 Research Method

The research method for this study is represented in the form of a flowchart, which outlines the significant steps taken to develop the FTC system and validate its performance. The flowchart guides the entire research process, beginning with the development of the FTC algorithm, followed by simulation and hardware testing, and concluding with the evaluation of results.

The flowchart represents the following key steps in the research process:

3.3.1 Fault-Tolerant Control Algorithm Development

The substeps involved in developing the FTC algorithm are critical to ensuring the reliable operation of the DAB converter, even in the presence of faults.

1. **Set Up Simulation Environment:** The simulation environment is prepared using PLECS, with the DAB converter model configured to simulate various OCF fault

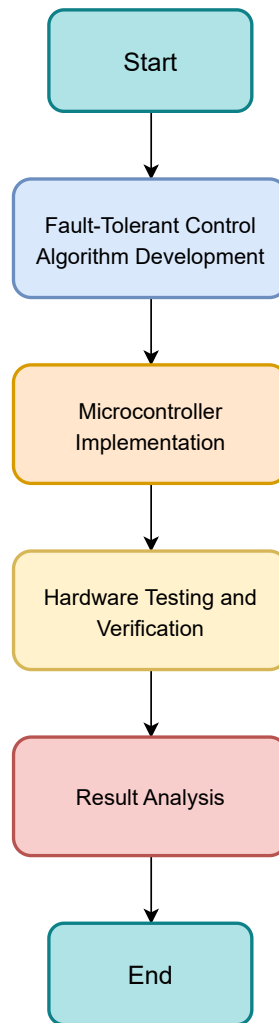


Figure 3.2. Flowchart of the Research

conditions. This setup provides a foundation for testing the control algorithm before hardware implementation.

2. **Develop Fault-Tolerant Control Algorithm:** A fault-tolerant control algorithm is designed to detect faults using a single current sensor. The algorithm adjusts the phase shifts between the primary and secondary sides of the converter to ensure fault tolerance, enabling continued operation during fault conditions.
3. **Implement Transition Logic:** Logic is created to ensure a smooth transition between normal and fault-tolerant operation modes. Upon detecting a fault, the system transitions to APS mode, which ensures system stability and prevents performance degradation.
4. **Test Fault Scenarios and Optimize Algorithm:** The control algorithm is tested under various OCF fault scenarios to assess fault recovery time, system stability, and efficiency. Based on the results, the algorithm is optimized to improve performance under fault conditions.
5. **Validate Algorithm Performance:** Finally, the algorithm's performance is validated in the simulation environment. The system's fault response is thoroughly

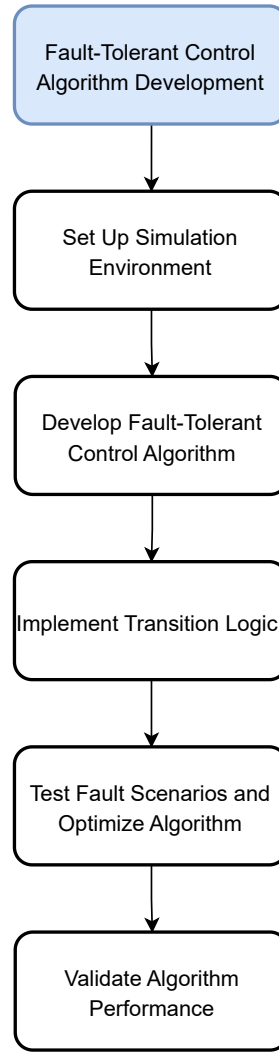


Figure 3.3. Flowchart of the Fault-Tolerant Control Algorithm

tested, and discrepancies between expected and actual behavior are addressed. Once the algorithm meets the desired performance criteria, it is validated for hardware implementation.

3.3.2 Microcontroller Implementation

This section outlines the steps for implementing the FTC system on the LAUNCHXL-F28379D microcontroller. It focuses on converting the control algorithm from simulation to a hardware-supported implementation and testing its operation with the hardware.

1. **Convert Algorithm for Microcontroller:** Initially developed in the simulation environment, the control algorithm is converted to a format compatible with the LAUNCHXL-F28379D microcontroller. The conversion involves adapting the algorithm for real-time operation and ensuring compatibility with the microcontroller's DSP core (TMS320F28379D). The adapted algorithm utilizes the DSP core's GPIO for PWM modulation, fault detection, and phase shift adjustments.

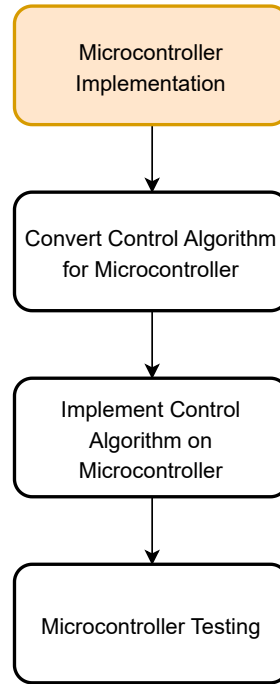


Figure 3.4. Flowchart of the Microcontroller Implementation

2. **Implement Control Algorithm on Microcontroller:** The converted control algorithm is implemented onto the LAUNCHXL-F28379D microcontroller. The implementation includes programming the microcontroller and ensuring proper interaction with the hardware components. The microcontroller executes the control algorithm and adjusts the converter's phase shifts based on fault detection. It also enables real-time communication between the microcontroller and the hardware.
3. **Microcontroller Testing:** After implementation, the microcontroller's response to normal and fault conditions is tested in a controlled environment. Its performance in adjusting phase shifts via PWM outputs and reading sensor data through ADCs is evaluated. It is ensured that the microcontroller effectively handles fault-tolerant operations before proceeding to real-world testing.

3.3.3 Hardware Testing and Verification

Once the hardware is implemented, it must undergo thorough testing to verify its performance under real-world conditions. The following substeps outline the hardware testing and verification phases for the fault-tolerant control system.

1. **Prepare Hardware Setup:** The hardware environment is prepared for fault testing, ensuring that the DAB converter and all associated components (e.g., sensors, power supply, microcontroller) are correctly configured and ready for simulating fault conditions.
2. **Test Hardware Under Fault Conditions:** Various fault conditions, such as open-circuit faults and faulty switches, are introduced to the hardware setup. The system's response to these faults is tested to ensure that the fault-tolerant control algorithm functions as expected. The algorithm ensures minimal performance

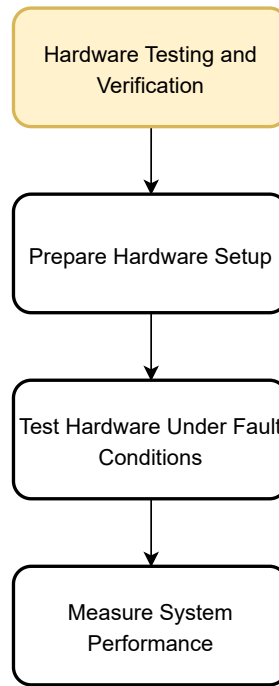


Figure 3.5. Flowchart of the Hardware Implementation

degradation during fault conditions.

3. **Measure System Performance:** Key performance metrics, such as fault recovery time, system efficiency, and stability, are measured during hardware testing. Data is collected to evaluate the real-world performance of the fault-tolerant system under fault conditions.

3.3.4 Result Analysis

The final step involves analyzing the simulation and hardware testing data. The system's performance in simulation is compared with its performance in real-world testing, discrepancies are identified, and conclusions are drawn regarding the effectiveness of the fault-tolerant control system. Additionally, potential improvements and future applications of the system are discussed. This analysis evaluates the system's performance and highlights areas for further research and enhancement.

CHAPTER IV

RESEARCH SCHEDULE

The research schedule for the upcoming study is outlined in Table 4.1. This table presents the planned activities and their corresponding timelines from mid-2025 to mid-2026, or almost one year. Each activity is carefully structured to ensure the smooth progression of the research. The structure covers the initial planning and algorithm testing through the final documentation and thesis writing stages. The table helps visualize the flow of tasks and deadlines, which is crucial for effective time management and project tracking throughout the research process.

Table 4.1. Research Activity Plan Schedule

No	Activity Description	Year 2025 month						Year 2026 month					
		7	8	9	10	11	12	1	2	3	4	5	6
1	Planning and Preparation												
2	Algorithm Testing and Refinement												
3	Microcontroller Implementation												
4	Hardware Testing and Verification												
5	Result Analysis												
6	Mid-Research Report Preparation												
7	International Paper Preparation												
8	International Paper Submission												
9	Documentation and Thesis Writing												
10	Pre-Pendadaran Preparation												
11	Pendadaran												
12	Yudisium												

REFERENCES

- [1] L. Li, G. Xu, D. Sha, Y. Liu, Y. Sun, and M. Su, "Review of Dual-Active-Bridge Converters With Topological Modifications," *IEEE Transactions on Power Electronics*, vol. 38, no. 7, pp. 9046–9076, Jul. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10075526>
- [2] D. Das, S. Mishra, and B. Singh, "Design Architecture for Continuous-Time Control of Dual Active Bridge Converter," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 9, no. 3, pp. 3287–3295, Jun. 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9109581>
- [3] J.-Y. Lee, H.-S. Kim, and J.-H. Jung, "Enhanced Dual-Active-Bridge DC–DC Converter for Balancing Bipolar Voltage Level of DC Distribution System," *IEEE Transactions on Industrial Electronics*, vol. 67, no. 12, pp. 10 399–10 409, Dec. 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/8936567>
- [4] Y. Yan, H. Bai, A. Foote, and W. Wang, "Securing Full-Power-Range Zero-Voltage Switching in Both Steady-State and Transient Operations for a Dual-Active-Bridge-Based Bidirectional Electric Vehicle Charger," *IEEE Transactions on Power Electronics*, vol. 35, no. 7, pp. 7506–7519, Jul. 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/8911503>
- [5] Y. Yan, H. Gui, and H. Bai, "Complete ZVS Analysis in Dual Active Bridge," *IEEE Transactions on Power Electronics*, vol. 36, no. 2, pp. 1247–1252, Feb. 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9146771>
- [6] T. Chaudhury and D. Kastha, "A High Gain Multiport DC–DC Converter for Integrating Energy Storage Devices to DC Microgrid," *IEEE Transactions on Power Electronics*, vol. 35, no. 10, pp. 10 501–10 514, Oct. 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/9023387>
- [7] E. L. Carvalho, L. H. Meneghetti, E. G. Carati, J. P. d. Costa, C. M. d. O. Stein, and R. Cardoso, "Asymmetrical Pulse-Width Modulation Strategy for Current-Fed Dual Active Bridge Bidirectional Isolated Converter Applied to Energy Storage Systems," *Energies*, vol. 13, no. 13, p. 3475, Jan. 2020, number: 13 Publisher: Multidisciplinary Digital Publishing Institute. [Online]. Available: <https://www.mdpi.com/1996-1073/13/13/3475>
- [8] G. Vacheva and N. Hinov, "Modelling of Dual Active Bridge Converter for Application in EVs Charging Station," in *2022 13th National Conference with International Participation (ELECTRONICA)*, May 2022, pp. 1–4. [Online]. Available: <https://ieeexplore.ieee.org/document/9874424>
- [9] S. K. Rastogi, S. S. Shah, B. N. Singh, and S. Bhattacharya, "Vector-Based Open-Circuit Fault Diagnosis Technique for a Three-Phase DAB Converter," *IEEE Transactions on Industrial Electronics*, vol. 71, no. 7, pp. 8207–8211, Jul. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10254283>
- [10] N. Zhao, J. Liu, Y. Shi, J. Yang, J. Zhang, and X. You, "Mode Analysis and Fault-Tolerant Method of Open-Circuit Fault for a Dual Active-Bridge DC–DC

- Converter,” *IEEE Transactions on Industrial Electronics*, vol. 67, no. 8, pp. 6916–6926, Aug. 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/8827590>
- [11] C. Song, A. Sangwongwanich, Y. Yang, and F. Blaabjerg, “Open-Circuit Fault Diagnosis and Tolerant Control for 2/3-Level DAB Converters,” *IEEE Transactions on Power Electronics*, vol. 38, no. 4, pp. 5392–5410, Apr. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10004016>
 - [12] Y. Chen and Y. Zhang, “Fault Characteristics and Riding-Through Methods of Dual Active Bridge Converter Under Short-Circuit of the Load,” *IEEE Transactions on Power Electronics*, vol. 36, no. 8, pp. 9578–9591, Aug. 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9335474>
 - [13] P. S. Bhakar and K. Jayaraman, “A New Fault-Tolerant Scheme for Switch Failures in Dual Active Bridge DC-DC Converter,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 10, no. 6, pp. 7627–7637, Dec. 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9780131>
 - [14] J. Le, Y. Guan, C. Liu, Y. Wang, and D. Xu, “An Improved Dual Active Bridge DC-DC Converter with Fault Tolerant Ability,” in *2023 IEEE 2nd International Power Electronics and Application Symposium (PEAS)*, Nov. 2023, pp. 361–366. [Online]. Available: <https://ieeexplore.ieee.org/document/10395015>
 - [15] P. Pal, M. Poshtan, A. R. Beig, R. K. Behera, K. Al Hosani, and U. R. Muduli, “Enhancing Resilience of DAB Converters with Fault-Tolerant Approach,” in *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Feb. 2024, pp. 453–460, iSSN: 2470-6647. [Online]. Available: <https://ieeexplore.ieee.org/document/10509040>
 - [16] S. S. Khan and H. Wen, “Universal Sensorless Open-Circuit Fault Detection and Isolation Method of Dual-Active-Bridge Converters With Low-Cost Diagnostic Circuit,” *IEEE Transactions on Power Electronics*, vol. 37, no. 11, pp. 13 652–13 667, Nov. 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9795237>
 - [17] D. Johannesson, M. Nawaz, S. Norrga, A. Hallén, and H.-P. Nee, “Static and Dynamic Performance Prediction of Ultrahigh-Voltage Silicon Carbide Insulated-Gate Bipolar Transistors,” *IEEE Transactions on Power Electronics*, vol. 36, no. 5, pp. 5874–5891, May 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9207967>
 - [18] C. Li, J. Sheng, and D. Dujic, “Reliable Gate Driving of SiC MOSFETs With Crosstalk Voltage Elimination and Two-Step Short-Circuit Protection,” *IEEE Transactions on Industrial Electronics*, vol. 70, no. 10, pp. 10 066–10 075, Oct. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/9967926>
 - [19] M. Picot-Digoix, F. Richardeau, J.-M. Blaquière, S. Vinnac, S. Azzopardi, and T.-L. Le, “Quasi-Flying Gate Concept Used for Short-Circuit Detection on SiC Power MOSFETs Based on a Dual-Port Gate Driver,” *IEEE Transactions on Power Electronics*, vol. 38, no. 6, pp. 6934–6938, Jun. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10075505>

- [20] R. Zhang, X. Li, J. Ding, S. Chen, H. Yang, and H. Guo, "Review of IGBT Intelligent Gate Drive and Protection Strategies," *IEEE Transactions on Power Electronics*, vol. 39, no. 6, pp. 7392–7403, Jun. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10460148>
- [21] D. Bisi, L. Nguyen, P. Zuk, A. Gokhale, K. Coffey, T. Liu, B. Cruse, T. Hosoda, M. Kamiyama, P. Parikh, and U. Mishra, "Short-Circuit Protection for GaN Power Devices with Integrated Current Limiter and Commercial Gate Driver," in *2022 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Mar. 2022, pp. 181–185, iSSN: 2470-6647. [Online]. Available: <https://ieeexplore.ieee.org/document/9773446>
- [22] T.-T. Le, T.-T. Pham, M.-K. Nguyen, T.-D. Duong, D.-D. Nguyen, M.-C. Dinh, and S. Choi, "Open-Circuit Fault-Tolerant Method for Three-Phase CF-DAB Converter With Auto-Balancing Control," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 4, pp. 3482–3493, Aug. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10551875>
- [23] T.-T. Le, M.-i. Nguyen, T.-D. Duong, C. Wang, and S. Choi, "Open-Circuit Fault-Tolerant Control for a Three-Phase Current-Fed Dual Active Bridge DC–DC Converter," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 2, pp. 1586–1596, Feb. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/9732277>
- [24] H. Shi, H. Wen, G. Chen, Q. Bu, G. Chu, and Y. Zhu, "Multiple-Fault-Tolerant Dual Active Bridge Converter for DC Distribution System," *IEEE Transactions on Power Electronics*, vol. 37, no. 2, pp. 1748–1760, Feb. 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9520252>
- [25] S. S. Khan, H. Wen, H. Shi, Y. Hu, L. Jiang, and G. Chen, "Multiple Open-Circuit Fault Detection and Isolation Using Universal Low-Cost Diagnosis Circuits for Reconfigurable Dual-Active-Bridge Converters," *IEEE Transactions on Power Electronics*, vol. 38, no. 5, pp. 6504–6521, May 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10008074>
- [26] S. K. Rastogi, S. S. Shah, B. N. Singh, and S. Bhattacharya, "Mode Analysis, Transformer Saturation, and Fault Diagnosis Technique for an Open-Circuit Fault in a Three-Phase DAB Converter," *IEEE Transactions on Power Electronics*, vol. 38, no. 6, pp. 7644–7660, Jun. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10035295>
- [27] —, "Mode Analysis and Identification Scheme of Open-Circuit Fault in a Three-phase DAB Converter," in *2021 IEEE Energy Conversion Congress and Exposition (ECCE)*, Oct. 2021, pp. 2762–2769, iSSN: 2329-3748. [Online]. Available: <https://ieeexplore.ieee.org/document/9595447>
- [28] P. Lyu, J. Huang, G. Chen, and W. Cui, "An Open-Circuit Fault-Tolerant Asymmetric Half-Bridge Dual-Output Converter With Topology Reconstruction," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 7, pp. 6754–6764, Jul. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/9889033>

- [29] Y. Wang, Y. Guan, M. Molinas, O. B. Fosso, W. Hu, and Y. Zhang, "Open-Circuit Switching Fault Analysis and Tolerant Strategy for Dual-Active-Bridge DC–DC Converter Considering Parasitic Parameters," *IEEE Transactions on Power Electronics*, vol. 37, no. 12, pp. 15 020–15 034, Dec. 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9841015>
- [30] F. Cai, M. Zhan, Q. Chai, and J. Jiang, "Fault Diagnosis of DAB Converters Based on ResNet With Adaptive Threshold Denoising," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–10, 2022. [Online]. Available: <https://ieeexplore.ieee.org/document/9798792>
- [31] W. Zhang, D. Xu, P. N. Enjeti, H. Li, J. T. Hawke, and H. S. Krishnamoorthy, "Survey on Fault-Tolerant Techniques for Power Electronic Converters," *IEEE Transactions on Power Electronics*, vol. 29, no. 12, pp. 6319–6331, Dec. 2014. [Online]. Available: <https://ieeexplore.ieee.org/document/6731601>
- [32] X. Pei, S. Nie, Y. Chen, and Y. Kang, "Open-Circuit Fault Diagnosis and Fault-Tolerant Strategies for Full-Bridge DC–DC Converters," *IEEE Transactions on Power Electronics*, vol. 27, no. 5, pp. 2550–2565, May 2012. [Online]. Available: <https://ieeexplore.ieee.org/document/6061967>
- [33] M. Syazwan, N. M. Hidayat, K. Naidu, M. Umair, M. A. N. Zelan, and N. H. Nik Ali, "Redundancy-Based Fault-Tolerant Control for Dual Active Bridge Converters," in *2024 IEEE 22nd Student Conference on Research and Development (SCORED)*, Dec. 2024, pp. 645–651, iSSN: 2643-2447. [Online]. Available: <https://ieeexplore.ieee.org/document/10872649>
- [34] Y. Pan, Y. Yang, J. He, A. Sangwongwanich, C. Zhang, Y. Liu, and F. Blaabjerg, "A Dual-Loop Control to Ensure Fast and Stable Fault-Tolerant Operation of Series Resonant DAB Converters," *IEEE Transactions on Power Electronics*, vol. 35, no. 10, pp. 10 994–11 012, Oct. 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/9005177>
- [35] Y. Zhang, N. Hou, Z. Wang, and Y. Li, "An Open-Circuit Fault-Tolerant Scheme for Single-Phase Current-Fed Dual-Active-Bridge DC/DC Converter," *IEEE Transactions on Power Electronics*, vol. 40, no. 3, pp. 3968–3979, Mar. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10755171>
- [36] D. Mou, Q. Luo, J. Li, Y. Wei, and J. Chen, "The Novel Single Phase-Shift Modulation Scheme for Dual Active Bridge Converter," in *2020 IEEE 9th International Power Electronics and Motion Control Conference (IPEMC2020-ECCE Asia)*, Nov. 2020, pp. 2686–2690. [Online]. Available: <https://ieeexplore.ieee.org/document/9368032>
- [37] V. K. Tiwari, M. A. Lowast, M. D. A. Alam, M. I. Siddiqui, and M. A. Alam, "Implementation of Single-Phase Shift (SPS) and Extended Phase Shift (EPS) for Dual Active Bridge (DAB)," *Indian Journal of Science and Technology*, vol. 17, no. 41, pp. 4262–4269, Nov. 2024, publisher: The Indian Society of Education and Environment. [Online]. Available: <https://indjst.org/>

- [38] F. Tlili and F. Bacha, “Single Phase Shift Control strategy of Isolated Single Phase Dual Active Bridge DC/DC Converter,” in *2023 9th International Conference on Control, Decision and Information Technologies (CoDIT)*, Jul. 2023, pp. 2176–2181, iSSN: 2576-3555. [Online]. Available: <https://ieeexplore.ieee.org/document/10284312>
- [39] S. S. Khan and H. Wen, “A Comprehensive Review of Fault Diagnosis and Tolerant Control in DC-DC Converters for DC Microgrids,” *IEEE Access*, vol. 9, pp. 80 100–80 127, 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9440977>
- [40] “LAUNCHXL-F28379D Overview,” Mar. 2019. [Online]. Available: https://www.ti.com/lit/ug/sprui77c/sprui77c.pdf?ts=1749514539543&ref_url=https%253A%252F%252Fwww.google.com%252F

APPENDIX

Proof of Submission

Magister TETI



Krishna Laksheta

Online

MAIN NAVIGATION

Home

Profile Mahasiswa

Daftar Dosen

Agenda Dosen

Statistik Bimbingan Dosen

Daftar Teman Seangkatan

RPKPS Matakuliah

JMFT Matakuliah

Pemohonan Surat Akademik

Dokumen & Surat Peringatan

TUGAS AKHIR (TESIS)

Syarat Ujian Pra/Pendadaran

Template Kelengkapan Tesis

Pendaftaran Proposal Tesis

Approval Pra/Pendadaran

Pendaftaran Pra/Pendadaran

LOG OUT

Log Out

Selamat Datang... Mahasiswa Prodi S2 Teknik Elektro dan S2 Teknologi Informatika

FORM PENDAFTARAN PROPOSAL TESIS

Form Pendaftaran

Cek Status Pendaftaran

Periode Pendaftaran :

APRIL 2025

Judul Proposal :

FAULT-TOLERANT CONTROL FOR SINGLE-PHASE DUAL ACTIVE BRIDGE USING ADJUSTED PHASE SHIFT MODE

Abstrak :

The dual active bridge (DAB) converter is a widely used power conversion topology known for its bidirectional power flow and galvanic isolation. However, its reliability is compromised when faults occur, especially open-circuit faults (OCFs) that can disrupt power transfer and damage components. While many fault-tolerant control (FTC) strategies exist, most require additional hardware, increasing complexity and cost, or involve computationally expensive techniques. This research proposes a fault-tolerant control system for a single-phase DAB converter that addresses open-circuit faults without requiring significant hardware modifications or reconfiguration. The system utilizes a single-phase shift (SPS) in the pre-fault mode, transitioning to an adjusted phase shift (APS) mode during fault conditions. It handles multiple faulty switches using only one current sensor on the primary side, reducing sensor requirements and simplifying the system design. The proposed approach minimizes computational power, offers cost-effectiveness, and allows easy integration with microcontrollers for hardware verification. This research contributes to simplifying fault-tolerant control in DAB converters while improving reliability and efficiency, making it a valuable solution for practical applications in electric vehicle charging, energy storage systems, and renewable energy integration.

Usulan Pembimbing I :

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Tanggal Daftar :

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Keterangan Status :

Data proposal sudah masuk. Silakan segera menghubungi calon dosen pembimbing 1 untuk proses kesiapan sebagai pembimbing 1

Edit Proposal

Abstrak

Unduh Proposal

Supervisor's Approval

RESEARCH PROPOSAL

**FAULT-TOLERANT CONTROL FOR SINGLE-PHASE
DUAL ACTIVE BRIDGE USING ADJUSTED PHASE
SHIFT MODE**



16/6/2025

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